

# Software Requirements Specification (SRS)

## AI CAREER COMPANION: RESUME BUILDER AND INTERVIEW PREPARATION SYSTEM

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|--------------------|-------------------------------|
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# Revision History

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| Version | Date | Description                                                                                                                           |
|---------|------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | —    | Initial SRS draft                                                                                                                     |
| 2.0     | —    | Enhanced with detailed functional decomposition, interface requirements, data model, use case specifications, and traceability matrix |

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# 1. Introduction

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## 1.1 Purpose

This document specifies the software requirements for the **AI Career Companion** platform, an AI-powered career preparation system designed to help students and fresh graduates build professional resumes, prepare for interviews, identify skill gaps, and improve overall employability through personalized, data-driven recommendations.

This SRS is intended for use by the development team, project guides/evaluators, and any stakeholders reviewing the system's scope and behavior. It defines *what* the system will do, not *how* it will be implemented internally.

## 1.2 Intended Audience

| Audience                | Purpose                                       |
|-------------------------|-----------------------------------------------|
| Developers              | Implementation reference                      |
| Project Guide/Evaluator | Academic assessment of scope and completeness |
| Testers/QA              | Basis for test case design                    |
| End Users (indirectly)  | Understanding of system capabilities          |

## 1.3 Scope

The system, referred to as **AI Career Companion**, is a web-based platform that enables users to:

- Register, log in, and manage a personal profile.
- Create, edit, and manage multiple resumes using structured templates.
- Analyze resumes for **ATS (Applicant Tracking System) compatibility** and receive a numeric score.
- Receive AI-generated suggestions to improve resume content and formatting.
- Practice **AI-generated mock interviews** based on selected job roles.
- Receive automated evaluation and feedback on interview answers.
- Identify **skill gaps** relative to a target job role and receive learning recommendations.
- Upload certificates, projects, and achievements to build a portfolio.
- Track career readiness progress over time via a personalized dashboard.
- (Administrator) Manage users, job role/skill taxonomies, and monitor platform usage.

### Out of Scope (for this version):

- Direct job application submission or integration with job portals.
- Payment processing / premium subscription billing.
- Video-based interview proctoring or facial expression analysis.
- Real-time human mentor/recruiter interaction.

## 1.4 Definitions, Acronyms, and Abbreviations

| Term                      | Definition                                                                                         |
|---------------------------|----------------------------------------------------------------------------------------------------|
| <b>AI</b>                 | Artificial Intelligence                                                                            |
| <b>ATS</b>                | Applicant Tracking System — software used by recruiters to filter resumes                          |
| <b>Resume Score</b>       | AI-generated numeric rating (e.g., 0–100) of resume quality and ATS-readiness                      |
| <b>Mock Interview</b>     | A simulated, AI-driven interview session with role-specific questions                              |
| <b>Skill Gap Analysis</b> | Comparative analysis between a user's current skills and the skills required for a target job role |
| <b>JWT</b>                | JSON Web Token, used for session authentication                                                    |
| <b>CRUD</b>               | Create, Read, Update, Delete                                                                       |
| <b>API</b>                | Application Programming Interface                                                                  |
| <b>UI/UX</b>              | User Interface / User Experience                                                                   |
| <b>SRS</b>                | Software Requirements Specification                                                                |

## 1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- OWASP Top 10 — Web Application Security Risks.
- REST API Design Guidelines.

## 1.6 Overview of Document

Section 2 describes the system in a general context. Section 3 details functional requirements grouped by module. Section 4 covers interfaces. Section 5 defines quality attributes. Sections 6–8 cover architecture, data, and use cases. Sections 9–11 cover constraints, traceability, and appendices.

## 2. Overall Description

### 2.1 Product Perspective

The AI Career Companion is a **new, self-contained web application** (not a modification of an existing system). It integrates with a third-party **AI/LLM API** (e.g., Gemini, OpenAI, or equivalent) for natural-language tasks such as resume feedback generation, interview question generation, and answer evaluation.

**High-level system composition:**

- Presentation Layer (Web UI — desktop & mobile responsive)
- Application/Business Logic Layer (Resume Engine, Interview Engine, Skill Gap Engine)
- AI Integration Layer (API gateway to external LLM service)
- Data Layer (relational/NoSQL database)
- Admin Panel (separate privileged interface)

### 2.2 Product Functions (Summary)

- User authentication and profile management
- Resume creation, editing, template selection, and export (PDF)
- ATS scoring and AI-based improvement suggestions
- Mock interview generation, response capture, and AI evaluation
- Skill gap analysis and learning resource recommendations
- Portfolio management (certificates, projects)
- Progress dashboard and historical reporting
- Administrative user, content, and role management

### 2.3 User Classes and Characteristics

| User Class           | Description                                                      | Technical Expertise | Frequency of Use                         |
|----------------------|------------------------------------------------------------------|---------------------|------------------------------------------|
| Student / Job Seeker | Primary end user: diploma/ engineering students, fresh graduates | Low–Medium          | Frequent (weekly/ daily during job prep) |
| Administrator        | Manages platform content, users, and monitors health             | Medium–High         | Occasional/Regular                       |

**Student/User capabilities:** Register/Login, Create & Edit Resumes, Download Resume, Attend Mock Interviews, View Feedback Reports, Manage Portfolio, View Skill Recommendations, Track Progress.

**Administrator capabilities:** Manage Users (activate/deactivate/delete), Manage Job Roles & Skill Taxonomy, Manage Resume Templates, View System/Usage Reports, Handle User Feedback/Support Tickets, Monitor AI Service Health.

## 2.4 Operating Environment

- **Client side:** Modern web browsers (Chrome, Firefox, Edge, Safari), desktop and mobile.
- **Server side:** Python (Flask/Django) backend hosted on a Linux server or cloud platform.
- **Database:** MySQL or Firebase (NoSQL).
- **External services:** AI/LLM API over HTTPS.

## 2.5 Design and Implementation Constraints

- Must use a REST API architecture between frontend and backend.
- AI-dependent features require an active internet connection and available third-party API quota.
- Resume export must produce standard, ATS-parseable PDF (no embedded images of text, proper text layers).
- Must comply with data protection practices for storing personal user data (resumes, certificates).

## 2.6 Assumptions and Dependencies

- Users have consistent internet connectivity while using AI-dependent features.
- The selected AI API provider maintains acceptable uptime and response latency.
- Users provide reasonably accurate profile, education, and experience data.
- The system depends on continued access to the third-party AI API; pricing/quota changes by the provider are an external risk.
- Browser must support modern JavaScript (ES6+) and HTML5 features.

# 3. System Features & Functional Requirements

## 3.1 Module: User Management & Authentication

| ID     | Requirement                                                                                                        | Priority |
|--------|--------------------------------------------------------------------------------------------------------------------|----------|
| FR-1.1 | The system shall allow a new user to register using email, password, and basic profile details.                    | High     |
| FR-1.2 | The system shall verify user email via a confirmation link or OTP before activating the account.                   | Medium   |
| FR-1.3 | The system shall allow registered users to log in securely using email/password or supported OAuth (e.g., Google). | High     |
| FR-1.4 | The system shall encrypt and securely store user passwords (hashed, never in plain text).                          | High     |
| FR-1.5 | The system shall allow users to reset forgotten passwords via a secure, time-limited link.                         | High     |
| FR-1.6 | The system shall allow users to update their profile information (name, education, contact, skills).               | Medium   |
| FR-1.7 | The system shall maintain session security using token-based authentication (e.g., JWT) with expiry.               | High     |

## 3.2 Module: Resume Builder

| ID     | Requirement                                                                                                                       | Priority |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-2.1 | The system shall allow users to create multiple resumes under a single account.                                                   | High     |
| FR-2.2 | The system shall provide at least 3 editable, ATS-friendly resume templates.                                                      | High     |
| FR-2.3 | The system shall allow users to add/edit/delete sections: personal info, education, skills, experience, projects, certifications. | High     |
| FR-2.4 | The system shall auto-save resume edits to prevent data loss.                                                                     | Medium   |
| FR-2.5 | The system shall allow users to export a completed resume as a PDF file.                                                          | High     |
| FR-2.6 | The system shall allow users to duplicate an existing resume as a starting point for a new one.                                   | Low      |
| FR-2.7 | The system shall validate mandatory fields (e.g., name, contact, at least one skill) before allowing export.                      | Medium   |

## 3.3 Module: ATS Analysis & AI Resume Feedback

| ID     | Requirement                                                                                                                              | Priority |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-3.1 | The system shall analyze a resume and generate an ATS compatibility score (0–100).                                                       | High     |
| FR-3.2 | The system shall identify missing or weak sections (e.g., no quantifiable achievements, missing keywords).                               | High     |
| FR-3.3 | The system shall suggest specific, actionable improvements using AI (e.g., rephrasing, keyword additions relevant to a target job role). | High     |
| FR-3.4 | The system shall allow users to compare resume scores before and after applying suggested edits.                                         | Medium   |
| FR-3.5 | The system shall complete resume analysis within 10 seconds under normal load (see NFR-1.2).                                             | High     |



### 3.4 Module: Mock Interview & AI Evaluation

| ID     | Requirement                                                                                             | Priority |
|--------|---------------------------------------------------------------------------------------------------------|----------|
| FR-4.1 | The system shall allow users to select a job role/domain for a mock interview session.                  | High     |
| FR-4.2 | The system shall generate role-specific interview questions (technical, behavioral, or mixed) using AI. | High     |
| FR-4.3 | The system shall capture user answers via text input (and optionally voice-to-text, if supported).      | Medium   |
| FR-4.4 | The system shall evaluate submitted answers using AI and generate a qualitative + quantitative score.   | High     |
| FR-4.5 | The system shall provide constructive feedback per answer (strengths, gaps, suggested improvement).     | High     |
| FR-4.6 | The system shall allow the user to repeat a mock interview session for the same or different job role.  | Medium   |
| FR-4.7 | The system shall generate a consolidated interview performance report at the end of a session.          | High     |

### 3.5 Module: Skill Gap Analysis & Recommendations

| ID     | Requirement                                                                                                       | Priority |
|--------|-------------------------------------------------------------------------------------------------------------------|----------|
| FR-5.1 | The system shall allow users to select or input a target career/job role.                                         | High     |
| FR-5.2 | The system shall compare the user's listed skills against the skill requirements of the target role.              | High     |
| FR-5.3 | The system shall display identified skill gaps in a clear, categorized format (e.g., must-have vs. good-to-have). | Medium   |
| FR-5.4 | The system shall recommend learning resources or courses to address identified skill gaps.                        | Medium   |

### 3.6 Module: Portfolio Management

| ID     | Requirement                                                                                  | Priority |
|--------|----------------------------------------------------------------------------------------------|----------|
| FR-6.1 | The system shall allow users to upload certificates (PDF/image formats).                     | Medium   |
| FR-6.2 | The system shall allow users to add project details (title, description, tech stack, links). | Medium   |
| FR-6.3 | The system shall allow users to organize portfolio items for inclusion in resumes.           | Low      |

### 3.7 Module: Progress Dashboard & Reporting

| ID     | Requirement                                                                         | Priority |
|--------|-------------------------------------------------------------------------------------|----------|
| FR-7.1 | The system shall maintain a history of resume scores over time per user.            | Medium   |
| FR-7.2 | The system shall maintain a history of mock interview scores and feedback per user. | Medium   |
| FR-7.3 | The system shall display an overall "career readiness" summary on a dashboard.      | Medium   |
| FR-7.4 | The system shall allow users to download/export their progress report.              | Low      |

### 3.8 Module: Administration

| ID     | Requirement                                                                                           | Priority |
|--------|-------------------------------------------------------------------------------------------------------|----------|
| FR-8.1 | The administrator shall be able to view, activate, deactivate, or delete user accounts.               | High     |
| FR-8.2 | The administrator shall be able to add/edit/remove job roles and associated skill taxonomies.         | High     |
| FR-8.3 | The administrator shall be able to manage available resume templates.                                 | Medium   |
| FR-8.4 | The administrator shall be able to view system usage reports (active users, sessions, feature usage). | Medium   |
| FR-8.5 | The administrator shall be able to review and respond to user-submitted feedback or issues.           | Low      |
| FR-8.6 | Only authenticated administrators with the correct role shall access the admin panel.                 | High     |

# 4. External Interface Requirements

## 4.1 User Interfaces

- Responsive web UI supporting desktop and mobile viewports.
- Consistent navigation: Dashboard, Resume Builder, Interview Prep, Skill Gap, Portfolio, Profile/Settings.
- Accessible forms with validation feedback (inline error messages).
- Loading indicators for AI-dependent operations (analysis, question generation, evaluation).

## 4.2 Hardware Interfaces

- No specialized hardware required beyond a standard client device (desktop/laptop/mobile) capable of running a modern browser.
- Optional microphone access if voice-based interview input is implemented.

## 4.3 Software Interfaces

| Interface                  | Purpose                                                            |
|----------------------------|--------------------------------------------------------------------|
| AI/LLM API (Gemini/OpenAI) | Resume feedback generation, question generation, answer evaluation |
| Database (MySQL/Firebase)  | Persistent storage of users, resumes, sessions, reports            |
| PDF Generation Library     | Resume export to PDF                                               |
| Email/OTP Service          | Account verification and password reset                            |
| Authentication Provider    | Third-party login (Google, etc.)                                   |

## 4.4 Communication Interfaces

- All client-server communication over HTTPS.
- RESTful JSON-based API between frontend and backend.
- Secure API key management for third-party AI service calls (server-side only, never exposed to client).

## 5. Non-Functional Requirements

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### 5.1 Performance Requirements

| ID      | Requirement                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------|
| NFR-1.1 | General page/API operations shall respond within 3 seconds under normal load.                       |
| NFR-1.2 | AI-based resume analysis shall complete within 10 seconds under normal load.                        |
| NFR-1.3 | The system shall support at least 100 concurrent users without significant performance degradation. |

### 5.2 Security Requirements

| ID      | Requirement                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------------------|
| NFR-2.1 | Passwords shall be stored using a strong hashing algorithm (e.g., bcrypt), never in plain text.              |
| NFR-2.2 | All data in transit shall be encrypted using HTTPS/TLS.                                                      |
| NFR-2.3 | Role-based access control shall restrict administrative functions to authorized admin accounts only.         |
| NFR-2.4 | The system shall protect against common web vulnerabilities (SQL injection, XSS, CSRF) per OWASP guidelines. |
| NFR-2.5 | User-uploaded files (certificates) shall be validated for type/size before storage.                          |

## 5.3 Reliability & Availability

| ID      | Requirement                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------|
| NFR-3.1 | The system shall target 99% uptime, excluding scheduled maintenance.                                             |
| NFR-3.2 | User data shall be backed up on a regular schedule (e.g., daily).                                                |
| NFR-3.3 | The system shall gracefully handle AI API failures/timeouts with a user-friendly error message and retry option. |

## 5.4 Usability

| ID      | Requirement                                                                               |
|---------|-------------------------------------------------------------------------------------------|
| NFR-4.1 | The UI shall follow a simple, consistent design usable by non-technical users.            |
| NFR-4.2 | The system shall be fully usable on both desktop and mobile browsers (responsive design). |
| NFR-4.3 | Error messages shall be clear, specific, and actionable.                                  |

## 5.5 Maintainability

| ID      | Requirement                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------------------------------|
| NFR-5.1 | The codebase shall follow modular architecture to allow independent updates to Resume, Interview, and Skill Gap modules. |
| NFR-5.2 | The AI integration layer shall be abstracted so the underlying AI provider can be swapped with minimal code change.      |

## 5.6 Scalability & Portability

| ID      | Requirement                                                                                      |
|---------|--------------------------------------------------------------------------------------------------|
| NFR-6.1 | The system architecture shall support horizontal scaling of backend services as user load grows. |
| NFR-6.2 | The application shall be deployable on common cloud platforms without OS-specific dependencies.  |

## 6. System Architecture (High-Level)

```
[ Client Browser (Web/Mobile) ]
    |
    v
[ Presentation Layer - Frontend (HTML/CSS/JS) ]
    |
    v (REST API / HTTPS)
[ Application Layer - Backend (Flask/Django) ]
  |--- Resume Management Module
  |--- Interview Management Module
  |--- Skill Gap Analysis Module
  |--- User & Admin Management Module
    |
    -----
    |               |
    v               v
[ AI Integration Layer ] [ Database Layer (MySQL/Firebase) ]
    |
    v
[ External AI/LLM API ]
```

### Key architectural notes:

- The AI Integration Layer isolates all third-party API calls, enabling provider swaps and centralized error handling/rate-limiting.
- The Admin Panel consumes the same backend API but is gated by role-based access control.

# 7. Data Requirements (Conceptual Data Model)

## 7.1 Core Entities

| Entity           | Key Attributes                                                                                    |
|------------------|---------------------------------------------------------------------------------------------------|
| User             | user_id, name, email, password_hash, education, created_at, role (student/admin)                  |
| Resume           | resume_id, user_id (FK), title, template_id, sections (JSON), ats_score, created_at, updated_at   |
| InterviewSession | session_id, user_id (FK), job_role, questions (JSON), answers (JSON), score, feedback, created_at |
| SkillGapReport   | report_id, user_id (FK), target_role, current_skills, missing_skills, recommendations, created_at |
| PortfolioItem    | item_id, user_id (FK), type (certificate/project), title, file_url/description, created_at        |
| JobRole          | role_id, role_name, required_skills (JSON), category                                              |
| AdminActionLog   | log_id, admin_id (FK), action_type, target_entity, timestamp                                      |

## 7.2 Key Relationships

- One **User** → many **Resumes**, **InterviewSessions**, **PortfolioItems**, **SkillGapReports** (1-to-many relationship).
- **JobRole** is referenced by **InterviewSession** and **SkillGapReport** for role-specific dynamic content generation.

# 8. Use Case Specifications

## 8.1 Use Case Summary Table

| Actor   | Use Case                                                                                                                                                                                                   |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Student | Register / Login, Create Resume, Edit Resume, Download Resume (PDF), Check ATS Score, Practice Mock Interview, Receive AI Feedback, Upload Certificates, View Skill Recommendations, Track Career Progress |
| Admin   | Manage Users, Manage Job Categories, View Reports, Monitor System                                                                                                                                          |

## 8.2 Detailed Use Case: Practice Mock Interview

|                       |                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>    | UC-06                                                                                                                                                                                                                                                                                                                                            |
| <b>Actor</b>          | Student/User                                                                                                                                                                                                                                                                                                                                     |
| <b>Preconditions</b>  | User is logged in; at least one job role is available in the system                                                                                                                                                                                                                                                                              |
| <b>Trigger</b>        | User selects "Start Mock Interview" and chooses a job role                                                                                                                                                                                                                                                                                       |
| <b>Main Flow</b>      | <ol style="list-style-type: none"><li>1. System generates role-specific questions via AI.</li><li>2. User answers each question (text input).</li><li>3. System submits answers to AI for evaluation.</li><li>4. System displays per-question feedback and an overall score.</li><li>5. System saves the session to interview history.</li></ol> |
| <b>Alternate Flow</b> | If AI service is unavailable, system displays an error and allows retry.                                                                                                                                                                                                                                                                         |
| <b>Postconditions</b> | Interview session and results are stored; dashboard updated with new performance data.                                                                                                                                                                                                                                                           |



## 8.3 Detailed Use Case: Check ATS Score

|                |                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID    | UC-05                                                                                                                                                                                |
| Actor          | Student/User                                                                                                                                                                         |
| Preconditions  | User has created at least one resume                                                                                                                                                 |
| Trigger        | User selects "Analyze Resume"                                                                                                                                                        |
| Main Flow      | 1. System sends resume content to AI analysis engine.<br>2. AI returns ATS score and improvement suggestions.<br>3. System displays score and suggestions to user within 10 seconds. |
| Alternate Flow | If analysis fails/times out, system notifies user and offers retry.                                                                                                                  |
| Postconditions | Score and suggestions are stored in resume history for progress tracking.                                                                                                            |

# 9. System Constraints

## 9.1 Hardware Requirements

- Minimum 4 GB RAM (client), dual-core processor
- Stable internet connection

## 9.2 Software Requirements

- OS: Windows/Linux (server); any OS with modern browser (client)
- Backend: Python (Flask or Django)
- Frontend: HTML, CSS, JavaScript
- Database: MySQL or Firebase
- AI API: Gemini, OpenAI, or equivalent LLM provider

## 9.3 Limitations

- AI-generated suggestions and evaluations are dependent on the quality and completeness of user-provided input.
- AI-powered features are unavailable without an active internet connection.
- Accuracy of ATS scoring and interview evaluation is bounded by the capabilities of the underlying third-party AI model and is not a guarantee of real-world recruiter outcomes.

# 10. Requirements Traceability Matrix (Sample)

| Requirement ID | Related Use Case | Priority | Module          |
|----------------|------------------|----------|-----------------|
| FR-1.1–FR-1.7  | UC-01            | High     | User Management |
| FR-2.1–FR-2.7  | UC-02, UC-03     | High     | Resume Builder  |
| FR-3.1–FR-3.5  | UC-05            | High     | ATS Analysis    |
| FR-4.1–FR-4.7  | UC-06            | High     | Mock Interview  |
| FR-5.1–FR-5.4  | UC-09            | Medium   | Skill Gap       |
| FR-6.1–FR-6.3  | UC-08            | Medium   | Portfolio       |
| FR-7.1–FR-7.4  | UC-10            | Medium   | Dashboard       |
| FR-8.1–FR-8.6  | UC-11–UC-14      | High     | Admin           |

# 11. Appendices

## 11.1 Future Enhancements

- Integration with job portals for direct application.
- Video/voice-based interview practice with tone and confidence analysis.
- Peer or mentor review of resumes.
- Gamified skill-building paths with certifications.

## 11.2 Glossary

See Section 1.4 for core definitions.

## 11.3 Assumptions Recap

- Users provide accurate and truthful profile/resume information.
- The chosen AI provider remains accessible and within acceptable cost/quota limits during the system's operational life.